

Programme	B. Sc. Computer Science				
Course Title	Computer Fundamentals with MS Excel,SPSS				
Type of Course	Minor				
Semester	I				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Fundamental Mathematics Concepts 2. Basic computer knowledge				
Course Summary	The course provides enough understanding of computer fundamentals, MS Excel, and SPSS. Students learn basic computing concepts, data entry, manipulation, and analysis in Excel and statistical analysis techniques using SPSS.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand fundamental concepts and skills essential for understanding and operating a computer system	U	C	Instructor-created exams / Seminar Presentation/ Instructor-created exams/ Quiz
CO2	Execute fundamental data input and manipulation tasks in MS Excel	C	P	Assignment / Instructor-created exams
CO3	Perform essential data input and manipulation activities within SPSS.	C	P	Assignment / Instructor-created exams

CO4	Implement Data analysis using SPSS	Ap	P	Hands-on practical sessions
CO5	Implement Data analysis using MS EXCEL	Ap	P	Hands-on practical sessions
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Introduction to computer system		12	19
	1	Features, Limitations, Types	1	
	2	Number systems and character representation, Binary arithmetic	2	
	3	Basic components of computer -	2	
	4	Computer software types,Utility Program, Operating systems functions and types	2	
	5	Input and output devices ,Primary memory and secondary storage	2	
	6	Overview of Emerging Technologies: cloud computing, big data, data mining, mobile computing and embedded systems	2	
	7	Use of Computers in Education and Research: Data analysis, Heterogeneous storage, e-Library, Google Scholar, Domain specific packages such as SPSS, MATLAB, Mathematica etc	1	
II	Introduction to Spread Sheet		10	18
	7	MS Excel - Creating & Editing Worksheet, Formatting and Essential Operations	2	
	8	Formulas and Functions, Charts	2	
	9	Advanced features: Vlookup,Hlookup, Index, Address, Match, Offset, Transpose, Conditional Formatting, Data Sorting and Filtering	3	

	10	Pivot table & Pivot Chart	2	
	11	Linking and Consolidation	1	
III	Introduction to SPSS		12	18
	12	Features – Data View – Variable View – Output Viewer Window – Syntax Editor Window -	2	
	13	Open data file , Save , import from other data source ,data entry , labelling for dummy numbers	2	
	14	Recode in to same variable, Recode in to different variable, Transpose of data, Insert variables and cases	2	
	15	Merge variables and cases, Split, Select cases, Compute total scores	2	
	16	Table looks – Changing column - font style and sizes	2	
	17	Diagrammatic representation	2	
IV	Data Analysis Using Ms Excel & SPSS		12	15
	18	Estimation of mean, median and mode- Standard deviation and coefficient of variation.	3	
	19	Descriptive statistics, Parametric tests t-test (paired or unpaired), ANOVA (one-way- two way)	3	
	20	Pearson rank correlation, Linear regression	2	
	21	Non parametric tests: Mann Whitney U test, Wilcoxon signed rank test .	2	
	22	Kruskall Wallis test ,Chi- Square test5x	2	

V	Hands-on Word Processor and Presentation Tool:		30	
	Practical Applications, Case Study and Course Project			
		EXCEL 1. Create a chart 2. Measures of Central Tendency & Descriptive Statistics 3. Parametric Tests T-Test 4. Correlation & Linear Regression 5. Chi- Square Test SPSS 6. Descriptive Statistics 7. Paired –Samples T Test 8. One-Way ANOVA 9. Correlation & Linear Regression 10. Chi- Square Test	20	
		Case study: Data analysis study on publically available biological data like bacterial growth analysis.		

Reference Books:

1. A. Goel, Computer Fundamentals, Pearson Education, 2010.
2. P. Aksoy, L. DeNardis, Introduction to Information Technology, Cengage Learning, 2006
3. P. K.Sinha, P. Sinha, Fundamentals of Computers, BPB Publishers, 2007
4. Excel Functions and Formulas Paperback by Bernd Held
5. Microsoft Excel 2010 Data Analysis and Business Modeling Paperback by Winsto
6. Jeremy J. Foster (2001). Data analysis using SPSS for windows. New edition. Versions 8-10. Sage publications. London.
7. Michael S. Louis – Beck (1995). Data analysis an introduction, Series: quantitative applications in 1

8. the social sciences. Sage, Publications. London.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	2	2	1	-	2	3						
CO 2	-	-	2	-	2	3						
CO 3	-	-	2	-	2	3						
CO 4	-	-	2	-	2	3						
CO 5	-	-	2	-	2	3						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓	✓	✓
CO 5	✓	✓	✓	✓